

Course: B. Tech. Branch: Information Technology Semester: V

Subject Code & Name: Programming in Java BTITOE504C

Max Marks: 60

Date: 08-02-2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) How java differs from C and C++?	L1 / CO1	6
B) What are Annotations? Explain built in annotations with program as an example @Override @inherited.	L3 / CO1	6
C) Explain the scope and life time of variables with an example.	L2 / CO1	6
Q.2 Solve Any Two of the following.		12
A) Explain the following: i) Use of this keyword ii) Garbage collection in java iii) finalize () method	L1 / CO2	6
B) Compare method overloading and method overriding with suitable example.	L2 / CO2	6
C) Define package. What are the steps involved in creating user defined package with a program?	L1/ CO2	6
Q. 3 Solve Any Two of the following.		12
A) Write a program for user defined exception that checks the internal and external marks if the internal marks are greater than 40 it raises the exception "internal marks are exceed", if the external marks are greater than 60 exception is raised and display the message the "external marks are exceed".	L2/ CO3	6
B) What is exception? Explain the different exception handling mechanism with an example.	L1/ CO2	6
C) What is Thread. Explain the Life Cycle of Thread? Write a program to implement Runnable Interface.	L2/ CO2	6

Q.4 Solve Any Two of the following.		12
A) What is difference between Swing and AWT in Java?	L1/ CO1	6
B) Write a swing program to accept the name and course details from user as input and display the concatenated string as output in a label.	L3/ CO3	6
C) Write a program to draw the circle, rectangle, oval, line using the methods of Graphics class and explain any two methods with syntax.	L3/ CO3	6
Q. 5 Solve Any Two of the following.		12
A) Draw and explain the hierarchy of Writer stream classes.	L1/ CO1	6
B) Design the java program to accept input from user and store the result in a file.	L3/ CO2	6
C) Explain the FileInputStream and FileOutputStream class with the methods.	L1/ CO1	6

***** End *****

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Supplementary Semester Examination-2023

Course: B. Tech.

Branch :Information Technology

Semester :V

Subject Code & Name: BTITOE504C_Programming in Java

Max Marks: 60

Date: 17/08/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(CO's)	Marks
Q.1 Solve Any Two of the following.		12
A) Distinguish between the following terms:	CO1	6
a) Objects and Classes		
b) Inheritance and Polymorphism		
B) Explain the process of compiling and running the java application with the example.	CO1	6
C) Write down the program to find the largest number among two numbers using ternary operator.	CO2	6
Q.2 Solve Any Two of the following.		12
A) Write down the difference between method overloading and method overriding.	CO2	6
B) Write a java program to perform simple calculator operations.	CO3	6
C) What is applet? Draw a human face using applet programming.	CO3	6
Q.3 Solve Any Two of the following.		12
A) Explain life cycle of thread.	CO2	6
B) Explain how thread class methods can move from one state to another using yield (), stop () and sleep ().	CO3	6
C) What is an exception? Explain the following	CO2	6
i) Try	ii) Catch	
iii) Throw		
iv) Throws	v) finally	

Q.4 Solve Any Two of the following.		12
A) Write a program to draw the circle, rectangle, oval, line using the methods of Graphics class.	CO3	6
B) Write down the java programs to draw polygon using drawLines(), drawPolygon() and fillPolygon().	CO3	6
C) What is difference between Swing and AWT in Java?	CO3	6

Q. 5 Solve Any Two of the following.		12
A) Explain hierarchy of Input stream Classes.	CO2	6
B) Write down java program to reading bytes from file.	CO2	6
C) a) Find the runtime error in the program given below:	CO1	6

```

class A{
    public static void main(String args[ ])
    {
        int var1 = 15;
        int var2 = 5;
        int var3 = 0;
        int ans1 = var1 / var2;
        int ans2 = var1 / var3;
        System.out.println("Division of va1" + " by var2 is: " + ans1);
        System.out.println("Division of va1" + " by var3 is: " + ans2);
    }
}

```

b) Find the compile time error in the program given below:

```

class B {
    public static void main(String args[ ])
    {
        String s = "Programming with Java";

        System.out.println("Welcome to " + s)
    }
}

```

***** END *****

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech.

Branch :Information Technology

Semester :V

Subject Code & Name: BTITOE504C-- Programming in Java

Max Marks: 60

Date: 08/01/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(CO's)	Marks
Q.1 Solve Any Two of the following.		12
A) Explain History and Evolution of Java?	CO1,2	6
B) Write down the program to illustration of math functions supported by java language.	CO1,2	6
C) Write a java program to display all the odd numbers between 1 to 30 using for loop & if statement.	CO1,2	6
Q.2 Solve Any Two of the following.		12
A) Define a class circle having data members pi and radius. Initialize and display values of data members also calculate area of circle and display it.	CO3	6
B) Explain life cycle of Applet.	CO1,2	6
C) What will be the Output of the below code and give the explanation:	CO1	6
<pre>A) public class A { public static void main(String[] args) { if (true) break; } }</pre>		
Options: A) Nothing B) Error		
<pre>B) public class A { public static void main(String[] args) { System.out.println('j' + 'a' + 'v' + 'a'); } }</pre>		
Options: A) java B) Something else (Other than simple concatenation)		

```

C) public class Demo{
    public static void main(String[] arr){
        Integer num1 = 100;
        Integer num2 = 100;
        Integer num3 = 500;
        Integer num4 = 500;

        if(num1==num2){
            System.out.println("num1 == num2");
        }
        else{
            System.out.println("num1 != num2");
        }
        if(num3 == num4){
            System.out.println("num3 == num4");
        }
        else{
            System.out.println("num3 != num4");
        }
    }
}

```

Options:

- A) num1 == num2
num3 == num4
- B) num1 == num2
num3 != num4
- C) num1 != num2
num3 == num4
- D) num1 != num2
num3 != num4

Q. 3	Solve Any Two of the following.		12
A)	Write down java program for how Thread class methods can move a thread from one state to another.	CO2,3	6
B)	Explain following clause w.r.t. exception handling : (i) try (ii) catch (iii) throw (iv) finally	CO2	6
C)	Write down and explain broadly classified errors with appropriate examples.	CO1,2	6
Q.4	Solve Any Two of the following.		12
A)	Differentiate between array and vector?	CO1	6
B)	Write down the code to draw the polygon using three different methods.	CO1,2	6
C)	Write down applet program to draw following shapes:	CO2	6
	a) Line		
	b) Rectangle		
	c) fillRectangle		
	d) Oval		

- e) fillArc
- f) fillRoundRectangle

Q. 5	Solve Any Two of the following.		12
A)	Write down classification of Java stream classes with appropriate hierarchy in detail.	CO1,2	6
B)	Write down and explain the program of copying bytes from one file to another.	CO3	6
C)	Write down a program to create a simple sequential student file using window frames, create text fields that receive information from the user through the keyboard and then writes the information to file. A record of information contains Roll number, Name and marks obtained by student in test. Also Add buttons ENTER and DONE.	CO1,2	6

***** End *****

Supplementary Summer Examination-2024

Course: B. Tech.

Branch : Information Technology

Semester :V

Subject Code & Name: BTITOE504C Programming in Java

Max Marks: 60

Date: 08/07/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(CO's)	Marks
Q.1 Solve Any Two of the following.		12
A) Write down the program to illustration of math functions supported by java language.	CO1	6
B) Write down program for reading data from keyboard using readLine().	CO1	6
C) Explain the process of compiling and running the java application with the example.	CO1	6
Q.2 Solve Any Two of the following.		12
A) Define a class circle having data members pi and radius. Initialize and display values of data members also calculate area of circle and display it.	CO2	6
B) Explain life cycle of Applet.	CO2	6
C) Write a Java program to create a class called Employee with methods called work() and getSalary(). Create a subclass called HRManager that overrides the work() method and adds a new method called addEmployee().	CO2	6
Q.3 Solve Any Two of the following.		12
A) Write down and explain broadly classified errors with appropriate examples.	CO2	6
B) Write down java program for how Thread class methods can move a thread from one state to another.	CO1	6
C) Write down java program to catch the exception using multiple catch blocks.	CO2	6
Q.4 Solve Any Two of the following.		12
A) Write a program to draw the circle, rectangle, oval, line using the methods of	CO3	6

Graphics class.

- | | | |
|--|-----|---|
| B) Differentiate between array and vector? | CO3 | 6 |
| C) Write down the java programs to draw polygon using drawLines(), drawPolygon() and fillPolygon(). | CO3 | 6 |

Q. 5 Solve Any Two of the following. 12

- | | | |
|--|-----|---|
| A) a) Find the runtime error in the program given below: | CO2 | 6 |
|--|-----|---|

```
class A{
    public static void main(String args[ ])
    {
        int var1 = 15;
        int var2 = 5;
        int var3 = 0;
        int ans1 = var1 / var2;
        int ans2 = var1 / var3;
        System.out.println("Division of va1" + " by var2 is: " + ans1);
        System.out.println("Division of va1" + " by var3 is: " + ans2);
    }
}
```

b) Find the compile time error in the program given below:

```
class B {
    public static void main(String args[ ])
    {
        String s = "Programming with Java";

        System.out.println("Welcome to " + s)
    }
}
```

- | | | |
|--|-----|---|
| B) Write down a program to create a simple sequential student file using window frames, create text fields that receive information from the user through the keyboard and then writes the information to file. A record of information contains Roll number, Name and marks obtained by student in test. Also Add buttons ENTER and DONE. | CO2 | 6 |
| C) How to handle primitive data types? And write down program to reading and writing primitive data. | CO2 | 6 |

***** END *****

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE			
Regular and Supplementary Winter 2024			
Course: B.Tech	Branch: Information Technology	Semester : V	
Subject Code & Name: BTITOE504C & Programming in Java			
Max Marks: 60	Date: 13/ 02 /2025	Duration: 3 Hr.	
Instructions to the Students: 1. Each question carries 12 marks. 2. Question No. 1 will be compulsory and include objective-type questions. 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 5. 4. The level of question/expected answer as per CBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 5. Use of non-programmable scientific calculators is allowed. 6. Assume suitable data wherever necessary and mention it clearly.			
		(CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Which of the following is the correct syntax of the for loop in Java? a) for (int i = 0; i < 10; i++) b) for (int i = 0; i <= 10; i--) c) for (int i = 0; i < 10; --i) d) for (i = 0; i < 10)	CO1	
2	What is the output of the following code? int x = 10; if (x > 5) { System.out.println("Greater"); } else { System.out.println("Smaller"); } a) Smaller b) Greater c) Error d) No output	CO1	
3	Which statement is used to skip the current iteration of a loop in Java? a) break b) stop c) continue d) exit	CO2	
4	Which of the following is true about Java? a) Java is a compiled and interpreted language. b) Java is a purely compiled language. c) Java is purely an interpreted language. d) Java is an assembly language.	CO2	
5	What is the correct syntax to catch an exception in Java? a) try { } catch(Exception e) { } b) try { } catch (Error e) { } c) try { } catch(Exception e) throw { }	CO2	

	d) try { } except(Exception e) { }		
6	What is the purpose of an applet in Java? a) To create graphical user interfaces. b) To handle files and directories. c) To manage database connections. d) To create small applications that can be embedded in web pages.	CO2	
	What is the purpose of the run() method in the Thread class in Java? a) To start the thread. b) To block the thread. c) To stop the thread. d) To define the thread's task.	CO1	
8	What is the life cycle of a thread in Java? a) New -> Ready -> Running -> Blocked -> Terminated b) Ready -> Running -> Blocked -> Terminated c) New -> Blocked -> Running -> Terminated d) New -> Running -> Terminated	CO1	
9	Which interface does a class need to implement to create a thread in Java using the "Runnable" approach? a) Thread b) Runnable c) Task d) Process	CO1	
10	Which of the following keywords is used to explicitly throw an exception in Java? a) throw b) throws c) exception d) try	CO2	
11	Which of the following is a feature of method overloading in Java? a) Methods with the same name but different parameter types. b) Methods with the same return type. c) Methods with the same name and parameters. d) Methods with different names.	CO2	
12	Which of the following is true about Java strings? a) Strings are mutable. b) Strings are immutable. c) Strings are not objects. d) Strings cannot be concatenated.	CO2	

Q.2	Solve the following.		12
A)	Describe the evolution of Java, starting from its inception to its current version. Highlight the significant updates in each version.	CO1	6
B)	Write a Java program to read n integers from the user, store them in an array, and calculate the following: Sum of all elements Average of the elements Maximum and minimum values in the array	CO2	6
Q.3	Solve the following.		12
A)	Write a program in Java to demonstrate method overloading with multiple constructors and method overriding with inheritance.	CO2	6
B)	Write an HTML file to embed a Java applet in a web page. The applet should display the message "Hello, Applet World!" on the screen. Also write down steps to run applet.	CO3	6
Q.4	Solve Any Two of the following.		12
A)	How do you create a thread in Java? What is the purpose of the Thread.setPriority() method, and how does thread priority work?	CO1	6
B)	What is the difference between errors and exceptions in Java? How can you handle different types of exceptions using multiple catch statements?	CO2	6
C)	Enlist and Explain different states in the life cycle of a thread in Java?	CO1	6
Q.5	Solve Any Two of the following.		12
A)	In the context of graphics programming, how does event handling work for interactive drawing applications? How can mouse events be used to create dynamic graphics?	CO3	6
B)	Explain how to draw circles, arcs, and ellipses using Java's Graphics class. How can you customize their appearance such as the start and extent of arcs?	CO3	6

C)	Write a program to display a bar chart representing monthly sales figures for a business. The chart should include labelled axes, bars scaled according to the sales data, and corresponding labels for each month. Months: { "January", "February", "March", "April", "May", "June" } Sales: { 200, 150, 300, 400, 350, 250 } (in units)	CO3	6
Q. 6	Solve Any Two of the following.		12
A)	What is the difference between a file and a stream in Java? Explain how Java handles files using streams.	CO2	6
B)	What is interactive input and output in Java, and how do you handle user input and output using streams?	CO3	6
C)	How do you use streams for reading and writing characters in Java? Provide an example of how to handle text files.	CO2	6
	*** End ***		

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